TRIBHUVAN UNIVERSITY

INSTITUTE OF SCIENCE AND TECHNOLOGY

AMRIT SCIENCE CAMPUS

Department of Computer Science and Information Technology

Project Report: Development of a Web-Based Yakthung Research Library Management System

This is a project work report submitted in partial fulfillment of the requirements for the Bachelor of Science in Computer Science and Information Technology (BSc. CSIT) degree third year sixth semester.

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Abstract

This report presents the development of a Library Management System using various software engineering concepts. The system aims to streamline library operations, including book management, member management, and transaction processing. The project follows the software engineering principles outlined in the course syllabus, incorporating requirements engineering, system modeling, design, implementation, testing, and project management.[2]

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1. Chapter 1: Introduction

1.1. Introduction

Libraries serve as vital hubs for knowledge and information dissemination in society. Efficient management of library resources, including books, journals, and other materials, is crucial for their effective operation.[5] Traditionally, many libraries relied on manual systems for managing their resources and operations. However, with the increasing volume of information and the need for efficient retrieval and management, automated Library Management Systems (LMS) have become indispensable.[7] The evolution from manual to automated systems has brought significant benefits, including improved organization, reduced manual effort, enhanced accuracy, and better user experience.[8] This project report details the development of a web-based Library Management System designed to streamline library operations and provide users with easy access to library resources. The system aims to automate key tasks such as user management, book cataloging, transaction handling, and report generation, thereby enhancing the overall efficiency of library management. This library is currently hosted on https://yrlms.ningsangjabegu.com.np/

1.2. Problem Statement

Manual library management often presents several challenges. These can include time-consuming processes for tasks like book borrowing and returning, difficulties in tracking overdue books, inefficiencies in generating reports, and potential for errors in maintaining records.[10] For library users, searching for books and managing their borrowing history can also be cumbersome in a manual system. The lack of a centralized, easily accessible system can lead to inefficiencies for both library staff and patrons.[11] This project addresses these challenges by developing a web-based Library Management System that automates these processes, providing a more efficient and user-friendly experience.

1.3. Objectives

The primary objectives of this project are to design and develop a fully functional web-based Library Management System with the features and functionalities outlined in the syllabus of this course. These objectives include:

a. To create a secure login and registration system for different user roles: Administrator, Author, and User.
b. To develop role-based dashboards providing relevant information and quick access to key functionalities.
c. To implement a comprehensive book management module allowing for adding, editing, deleting, and searching of books.
d. To create a category management system for organizing books into different genres or subjects.
e. To develop user and team management modules for maintaining records of library members and system users.
f. To implement a transaction management system for tracking book borrowing and returning activities.
g. To generate various reports providing insights into library operations, such as book statistics, member activity, and overdue books.
h. To include settings and notification functionalities for system customization and user updates.

1.4. Scope and Limitation

The scope of this project encompasses the development of a web-based application with the features and functionalities detailed in the user query. This includes user authentication, role-based access control, book and category management, user and team management, transaction tracking, report generation, system settings, and notifications. The system is designed to be accessible through web browsers, providing a centralized platform for library management. However, certain aspects may be considered limitations for this version (version 0.2), such as advanced integration with other library systems or functionalities beyond those explicitly mentioned in the user query. Future iterations could potentially address these limitations.

1.5. Development Methodology

An Agile software development methodology was adopted for this project. Agile methodologies emphasize iterative development, collaboration, and flexibility in response to changing requirements.[2] This approach allows for breaking down the project into smaller, manageable iterations, with regular feedback and adjustments throughout the development process. The focus on working software and customer collaboration aligns well with the need to develop a functional and user-centric Library Management System.[16]

1.6. Report Organization

This project report is organized into six chapters, including this introduction. Chapter 2 provides a background study of Library Management Systems and reviews relevant literature in the field. Chapter 3 details the system analysis phase, including requirement analysis, feasibility analysis, and system modeling using UML diagrams. Chapter 4 focuses on the system design, covering the architecture, database design, interface design, and security considerations. Chapter 5 describes the implementation process, the tools used, and the testing strategies employed, along with the analysis of the test results. Finally, Chapter 6 presents the conclusion of the project and offers recommendations for future enhancements. The report also includes a list of references and appendices containing supplementary materials.

2. Chapter 2: Background Study and Literature Review

2.1. Background Study

The concept of a library has evolved significantly over time, from repositories of physical scrolls and manuscripts to modern centers of information that include digital resources. Regardless of their form, libraries play a crucial role in preserving and providing access to knowledge.[6] The effective management of these resources is paramount to the success of any library. Traditional manual systems, while serving their purpose for centuries, often struggle to cope with the demands of modern libraries, which typically manage large collections and serve a diverse user base.[6] The advent of computer technology presented an opportunity to automate library operations, leading to the development of Library Management Systems.[7] These systems are designed to streamline various library tasks, including cataloging, circulation, user management, and reporting.[5] Modern LMS often incorporate web-based interfaces, allowing for remote access and enhanced user interaction. Key features commonly found in such systems include user authentication with different roles and permissions, comprehensive book cataloging and search functionalities, efficient transaction management for borrowing and returning materials, and the ability to generate insightful reports on library activities.

2.2. Literature Review

The development of Library Management Systems has been a subject of extensive research and practical implementation. Several studies highlight the benefits of automating library operations. For instance, a study by Ayo et al. (2023) emphasizes the importance of needs assessment in the design and development of an integrated library management system, focusing on factors like technology infrastructure and user readiness.[18] Their research underscores the necessity of understanding user needs and incorporating the latest technological advancements to create effective LMS. Similarly, Tirupathi (2014) traces the evolution of library management systems from the mid-1950s, noting their increasing sophistication and integration of various library functions.[7]

Academic literature also provides insights into the architectural design and functionalities of LMS. The design often includes modules for user management, book cataloging, circulation control, and report generation.[21] The use of web technologies allows for broader accessibility and user interaction, as discussed in research on web application development best practices.[22] These best practices emphasize the importance of planning, choosing the right technology stack, creating an attractive and intuitive user interface, ensuring speed and performance, building for scalability, and implementing robust security measures.

Database design is another critical aspect of LMS development. Principles of database design, including data integrity, scalability, flexibility, and performance, are essential for creating a robust and efficient system.[26] Normalization techniques are employed to reduce data redundancy and improve data integrity, ensuring the reliability of the information stored in the database.[30]

Software engineering principles guide the entire development process, from requirements engineering to testing and deployment.[16] Requirements engineering focuses on eliciting, documenting, and managing the needs of the stakeholders.[46] System analysis and design methodologies provide frameworks for modeling the system's structure and behavior.[50] Testing methodologies ensure the quality and reliability of the developed software.[58] Project management principles are applied to plan, execute, and monitor the software development process effectively.[62]

The developed web-based Library Management System integrates these principles and best practices to create an efficient and user-friendly solution for managing library resources and operations.

3. Chapter 3: System Analysis

3.1. System Analysis

3.1.1. Requirement Analysis

Functional Requirements. The web-based Library Management System is designed to provide a comprehensive set of functionalities to manage library operations efficiently. The functional requirements are as follows:

User Authentication. The system must allow new users to register and existing users to log in based on their roles: Administrator, Author, and User. Each role will have different levels of access and functionalities within the system.

Role-Based Dashboards. Upon successful login, users will be redirected to their respective dashboards. The Administrator's dashboard will display all available tabs/options/sidebars (Dashboard, Books, Categories, Users, Teams, Transactions, Reports, Users). The Author's dashboard will have limited options (Dashboard, Books, Categories), and the User's view will be even more restricted (Dashboard, Books).

Dashboard Overview. The dashboard will provide a statistical overview of the library's current status, including the 'Total Books', 'Total Members', 'Transactions', and 'Overdue Books'. It will also display brief details on 'Recent Activities', a pie chart illustrating 'Book Categories' distribution, and a list of 'Popular Books' with their authors and categories. Quick links will be available for frequently performed actions such as 'Add New Book', 'Add New Member', 'New Transaction', and 'View Overdues'. A search box will enable users to search for books and authors.

Book Management. This module will provide a tabular listing of all books in the library, with headings for 'Title', 'Author', 'Category', 'Status' (Checked Out or Available), 'Borrower', 'Due Date', and 'Actions' (edit and delete buttons). A search box will allow users to search for specific books. Functionality to add new books and export the book list in PDF format will also be included.

Category Management. This section will feature a search box for categories and a tabulated display of existing categories with headings for 'Category Name', 'Book Count', and 'Actions' (edit and delete icons).

User (Member) Management. This module will list all library members in a tabular format with headings for 'Name', 'Email', 'Status' (active or inactive), 'Books' (currently borrowed), and 'Action'. A search box will be available for searching members, and buttons to add new users and export the member list in PDF format will be provided.

Team Management, Similar to user management, this section will allow for managing system users with different roles. It will include a search box, a table with headings for 'Name', 'Email', 'Role', 'Join Date', 'Status', and an 'Edit' option. Functionality to add new team members and export the team list will also be available.

Transaction Management. This module will allow library staff to record and manage borrowing and returning transactions. It will include an 'export' button, an 'add new transaction' button, a search box, and a table displaying 'Book', 'Member', 'Type' (Borrow/Return), 'Date', 'DueDate', 'Return Date', and 'Status' of each transaction.

Report Generation. This section will provide various reports on library activities. 'General Statistics' will show the 'Total Book' count, 'Total Member' count, 'Books Checked Out', 'Overdue Books', and a monthly activity bar graph. The 'Book Report' section will include a pie chart of 'Book Categories Distribution' and a timeline chart of 'Most Popular

Books'. 'Member Reports' will feature a pie chart of 'Member Activities' and a timeline of 'Most Active Members', along with a section for overdue books.

User Management (All Types). This section will list all types of system users ('Administrator', 'Author', 'User') with an 'Add User' button, a 'search box', and a table with headings for 'Name', 'Email', 'Role', 'Join Date', 'Status', and an 'Edit' option.

Settings. The system will include a 'Settings' section with subsections for 'General', 'Account', 'Notification', and 'About'. Each subsection will contain relevant information and input boxes to update the respective settings.

Logout. A 'Logout' button will be available to allow users to securely log out of the system and return to the login page.

Notifications. A notification icon will display the number of unread notifications. Clicking the icon will show a dropdown of recent notifications, and a 'View all notifications' option will lead to a page listing all notifications with brief details.

Profile Management. Clicking on the user's profile will trigger a dropdown menu with options for 'My Account' (containing 'Profile' and 'Settings'), the system version (0.2), and 'Logout'.

Non-Functional Requirements. In addition to the functional requirements, the Library Management System must also meet the following non-functional requirements:

Performance. The system should be responsive and load pages within an acceptable timeframe (e.g., under 3 seconds) to ensure a smooth user experience.[22]

Security. User data, including login credentials and personal information, must be securely stored and protected from unauthorized access. Secure coding practices should be implemented to prevent common web vulnerabilities.[23]

Usability**.** The system should have an intuitive and user-friendly interface that is easy to navigate for users with varying levels of technical expertise.[22] Clear instructions and feedback should be provided to guide users.

Reliability. The system should be stable and operate without frequent errors or crashes. Data integrity must be maintained to ensure the accuracy and consistency of library records.

Scalability**.** The system should be designed to handle a growing number of users, books, and transactions without significant degradation in performance.[22] The database architecture should be scalable to accommodate future growth.

Maintainability**.** The codebase should be well-structured, documented, and easy to understand to facilitate future maintenance, updates, and enhancements.

Accessibility. The system should strive to be accessible to users with disabilities, adhering to web accessibility guidelines where possible.

3.1.2. Feasibility Analysis

Technical Feasibility. The development of a web-based Library Management System is technically feasible. The required technologies, including web servers, database management systems, and programming languages, are readily available and well-established.[76] The team possesses the necessary skills in web application development and database design to undertake this project.

Operational Feasibility. The system is designed to integrate smoothly with existing library operations. The role-based access control ensures that different categories of library staff and users have access only to the functionalities relevant to their roles. The intuitive user interface will facilitate easy adoption and effective use of the system by library staff and patrons.[76]

Economic Feasibility. The economic feasibility of the project is positive. While there will be initial costs associated with development and deployment, the automation of library operations will lead to long-term cost savings through reduced manual effort, improved efficiency, and better resource management.[76] The system can potentially reduce the need for additional staff to handle routine tasks.

Schedule Feasibility. The project can be completed within a reasonable timeframe by following an Agile development approach with iterative planning and execution. A Gantt chart outlining the project timeline, including phases for analysis, design, implementation, testing, and deployment, will be developed to ensure adherence to the schedule.[4]

3.1.3. Analysis (Object Oriented Approach)

Object Modeling using Class and Object Diagrams. This class diagram illustrates the main entities in the Library Management System and their relationships. The User class represents library members and system users, with attributes for identification, contact information, role, and status. The Book class stores information about each book, including its title, author, category, and availability status. The Category class organizes books into different categories. The Transaction class records details of book borrowing and returning activities, linking books and users. The Report class represents the various reports that can be generated by the system.

Dynamic Modeling using State and Sequence Diagrams. ***(State Diagram for Book).*** A book can be in one of two states: 'Available' or 'Checked Out'. When a book is added to the library, its initial state is 'Available'. When a user borrows the book, its state changes to 'Checked Out'. Upon returning the book, its state reverts to 'Available'.

(Sequence Diagram for Borrow Book)

1. User selects a book to borrow.
2. System checks the availability of the book.
3. If available, system prompts for user confirmation.
4. User confirms the borrowing action.
5. System creates a new transaction record with the book ID, user ID, borrow date, and due date.
6. System updates the book status to 'Checked Out'.
7. System displays a confirmation message to the user.

Process Modeling using Activity Diagrams.

(Activity Diagram for Add New Book).

1. Administrator navigates to the 'Books' section.
2. Administrator clicks on the 'Add New Book' button.
3. System displays a form for entering book details (Title, Author, Category, ISBN, etc.).
4. Administrator enters the required book information.
5. Administrator clicks the 'Save' button.
6. System validates the entered data.
7. If valid, the system saves the book information to the database.
8. System displays a success message.

   If invalid, the system displays error messages indicating the issues.

4. Chapter 4: System Design

4.1. Design (Object Oriented Approach)

The system design phase builds upon the analysis phase, providing a detailed blueprint for the development of the Library Management System.

4.1.1. Refinement of UML Diagrams

The UML diagrams created during the analysis phase will be further refined to include more specific details about attributes, operations, and relationships. For example, the User class will include specific methods for different roles, and the Book class might include methods for updating its status. Sequence diagrams will be expanded to cover more complex interactions, such as handling overdue books or placing holds.

Component Diagrams. This component diagram shows the high-level components of the system. The 'User Interface' component handles user interactions and presents information. The 'Business Logic' component contains the core functionalities and rules of the system. The 'Database' component is responsible for storing and managing the system's data. The 'Reporting Module' generates various reports based on the data stored in the database.

Deployment Diagrams. The deployment diagram illustrates the physical deployment of the system. The 'User Interface' and 'Business Logic' components reside on the 'Web Server'. The 'Database' ('LibraryDB') is located on a separate 'Database Server'. Users interact with the system through their 'Client Browser' via HTTP or HTTPS. The web server communicates with the database server using protocols like JDBC or ODBC.

Database Design. The database schema for the Library Management System is designed based on the class diagram and normalized to the Third Normal Form (3NF) to minimize redundancy and ensure data integrity.

Database Schema.

Table Name	**Column Name**	**Data Type**	**Primary Key (Yes/No)**	**Foreign Key (Yes/No)**	**References**
Users	userID	INT	Yes	No	
	name	VARCHAR(255)	No	No	
	email	VARCHAR(255)	No	No	
	password	VARCHAR(255)	No	No	
	role	VARCHAR(50)	No	No	
	joinDate	DATETIME	No	No	
	status	VARCHAR(20)	No	No	
Books	bookID	INT	Yes	No	
	title	VARCHAR(255)	No	No	

	author	VARCHAR(255)	No	No	
	categoryID	INT	No	Yes	Categories(categoryID)
	ISBN	VARCHAR(20)	No	No	
	status	VARCHAR(20)	No	No	
	publicationDate	DATE	No	No	
Categories	categoryID	INT	Yes	No	
	categoryName	VARCHAR(255)	No	No	
Transactions	transactionID	INT	Yes	No	
	bookID	INT	No	Yes	Books(bookID)
	userID	INT	No	Yes	Users(userID)
	type	VARCHAR(10)	No	No	

	transactionDate	DATETIME	No	No	
	dueDate	DATE	No	No	
	returnDate	DATE	No	No	
	status	VARCHAR(20)	No	No	

This schema defines the tables necessary to store information about users, books, categories, and transactions. Primary keys uniquely identify each record in a table, while foreign keys establish relationships between tables, ensuring data consistency.

Interface Design. The user interface is designed with a focus on simplicity and ease of use, following best practices for web application development.[22] Key interfaces will include:

Login Page. A simple form for users to enter their email and password.

Dashboards. Role-specific dashboards providing an overview of relevant information and quick links to key features. Screenshots of the Administrator, Author, and User dashboards will be included in the appendices (Figure 4.1, Figure 4.2, Figure 4.3).

Book Listing Page. A tabular display of books with search and filtering options. A screenshot of the book listing page will be included in the appendices (Figure 4.4).

Add/Edit Book Form. A user-friendly form for adding new books or editing existing book details. A screenshot of the add book form will be included in the appendices (Figure

4.5).

Transaction Form. A form for recording new borrowing and returning transactions. A screenshot of the transaction form will be included in the appendices (Figure 4.6).

Report Pages. Display of various reports, including statistics, charts, and tables. Screenshots of the general statistics report and book category distribution chart will be included in the appendices (Figure 4.7, Figure 4.8).

Security Considerations. Security is integrated into the design of the Library Management System.[23] Measures include:

Password Hashing. User passwords will be hashed using a strong hashing algorithm before being stored in the database.

Input Validation. All user inputs will be validated on both the client-side and server-side to prevent malicious data injection.

Role-Based Access Control Access to system functionalities will be strictly controlled based on the user's role.

Secure Connections. HTTPS will be used to encrypt communication between the client and the server, protecting sensitive data during transmission.

Protection Against Common Vulnerabilities. Design patterns and secure coding practices will be employed to mitigate risks such as Cross-Site Scripting (XSS) and SQL Injection.

4.2. Algorithm Details

4.2.1. Book Search Algorithm

The system will implement a search algorithm that allows users to search for books based on keywords in the title, author, or ISBN. The algorithm will likely involve querying the Books table in the database using the provided keywords. A full-text search index on the relevant columns can be used to improve search performance.

4.2.2. Transaction Management Algorithm

When a user requests to borrow a book, the system will first check the book's status in the Books table. If the status is 'Available', a new record will be added to the Transactions table with the book ID, user ID, current date as the transaction date, and a calculated due date (e.g., based on a predefined loan period). The book's status in the Books table will then be updated to 'Checked Out'. When a book is returned, the system will update the corresponding transaction record with the return date and change the book's status back to 'Available'.

4.2.3. Popular Books Report Algorithm

To generate the 'Most Popular Books' report, the system will query the Transactions table to count the number of times each book has been borrowed within a specific time period (e.g., the last month or year). The results will be sorted in descending order of the borrow count, and the top N books will be displayed in the report.

5. Chapter 5: Implementation and Testing

5.1. Implementation

5.1.1. Tools Used

The following tools and technologies were used for the development of the Library Management System:

Programming Language. React with TypeScript was chosen as the primary programming language for its suitability for web development and its extensive ecosystem of frameworks and libraries.

Web Framework. React with TypeScript, a popular PHP framework, was used to streamline the development process by providing a robust structure, built-in features, and an elegant syntax.

Database Management System. MySQL was selected as the relational database management system for storing and managing the system's data due to its reliability, performance, and wide adoption in web applications.

Front-End Technologies. React with TypeScript, Node JS were used for developing the user interface, ensuring a dynamic and interactive user experience. Bootstrap, a CSS framework, was utilized for creating a responsive and visually appealing design.

Version Control System. Git was used for version control to track changes to the codebase and facilitate collaboration among team members. GitHub served as the remote repository for storing the project code.

Integrated Development Environment (IDE). Visual Studio Code (VS Code) was used as the IDE for writing and debugging the code due to its features, extensions, and ease of use.

The selection of these tools was based on their suitability for the project requirements, the team's familiarity with them, and their proven track record in developing web applications.[2]

5.1.2. Implementation Details of Modules

User Authentication Module. This module implements the login and registration functionalities for different user roles. Laravel's built-in authentication features were utilized to handle user registration, login, and password management. Middleware was implemented to enforce role-based access control, ensuring that users can only access the parts of the system relevant to their roles.

Book Management Module. This module allows administrators and authors to add, edit, and delete book records in the Books table. Forms were created using Laravel's Blade templating engine for data input. Data validation was implemented to ensure the integrity of the book information. The module also includes functionality for searching books based on various criteria using database queries. The export to PDF functionality was implemented using a PHP library for PDF generation.

Transaction Management Module. This module handles the borrowing and returning of books. When a borrow transaction is initiated, a new record is created in the Transactions table, linking the user and the book. The book's status in the Books table is updated accordingly. Similarly, when a book is returned, the corresponding transaction record is updated with the return date, and the book's status is changed back to 'Available'.

Report Generation Module. This module generates various reports based on the data in the database. Database queries were used to retrieve the necessary information for each report. For example, the 'Total Books' count is obtained by querying the number of records in the Books table. Charts and graphs were generated using JavaScript libraries based on the retrieved data to provide visual representations of the information.

Snippet of User Authentication Controller Code.

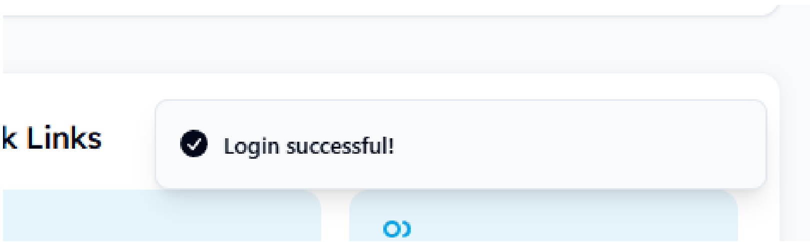
5.2. Testing

5.2.1. Test Cases for Unit Testing

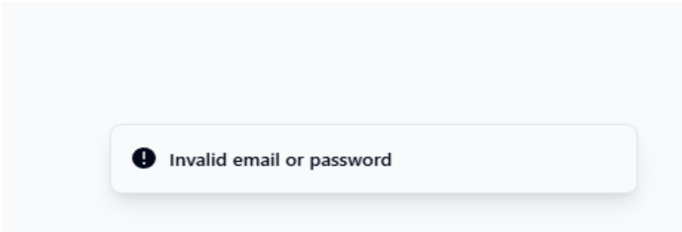
Unit testing was performed on individual components and functions of the system to ensure they function correctly in isolation.[2] Examples of unit test cases include:

Testing User Registration.

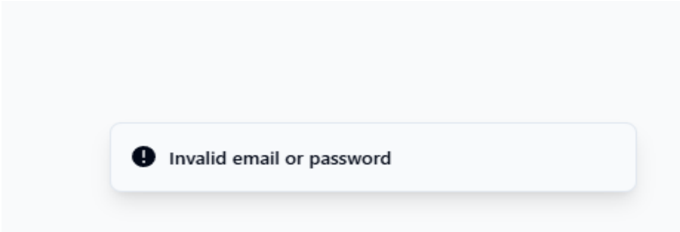
Test case 1. Verify that a new user can be successfully registered with valid credentials.



Test case 2. Verify that registration fails if an email address already exists in the database.

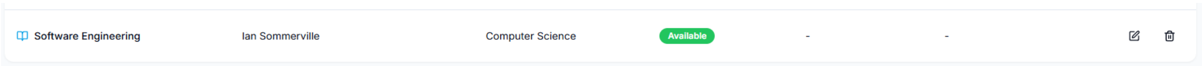


Test case 3. Verify that registration fails if the password does not meet the minimum length requirements.

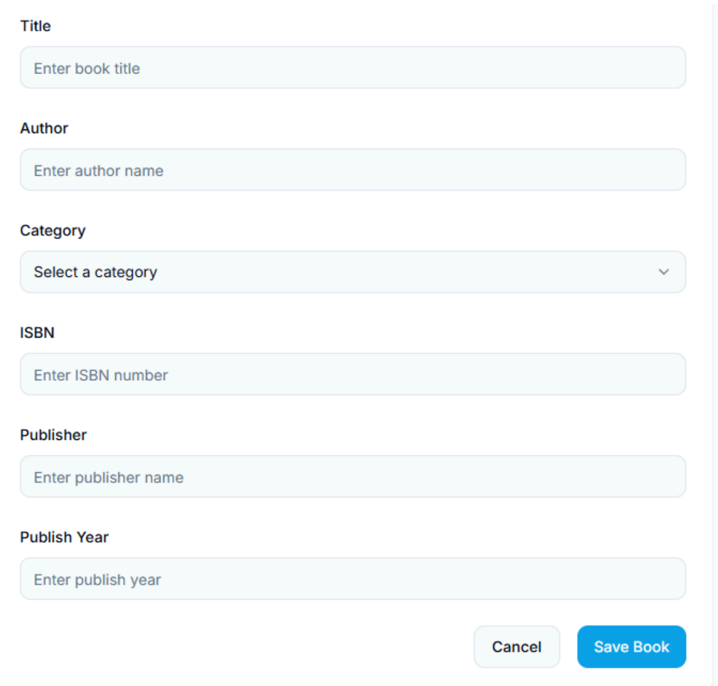


Testing Book Creation.

Test case 1. Verify that a new book can be added to the database with valid details.

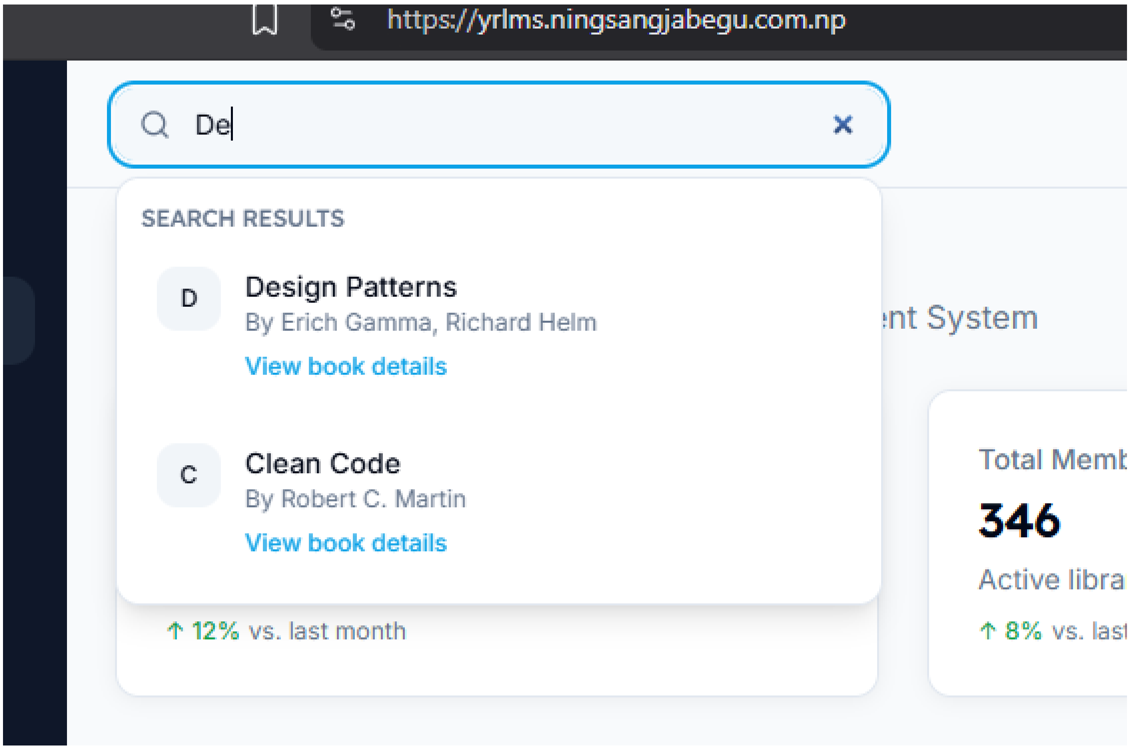


Test case 2. Verify that adding a book fails if required fields (e.g., title, author) are missing.

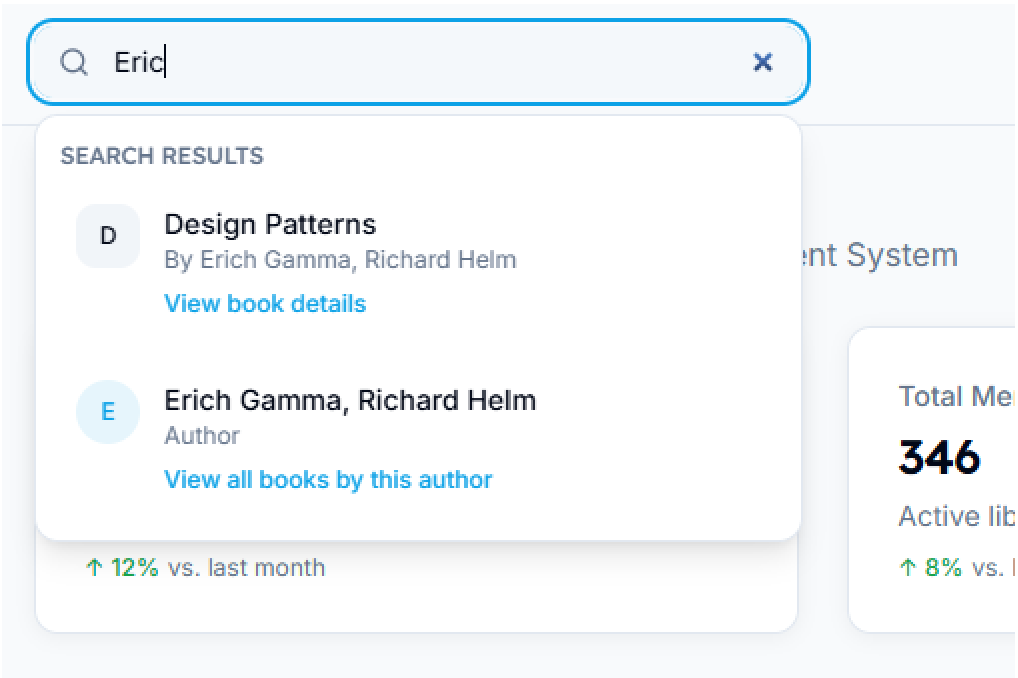


Testing Book Search.

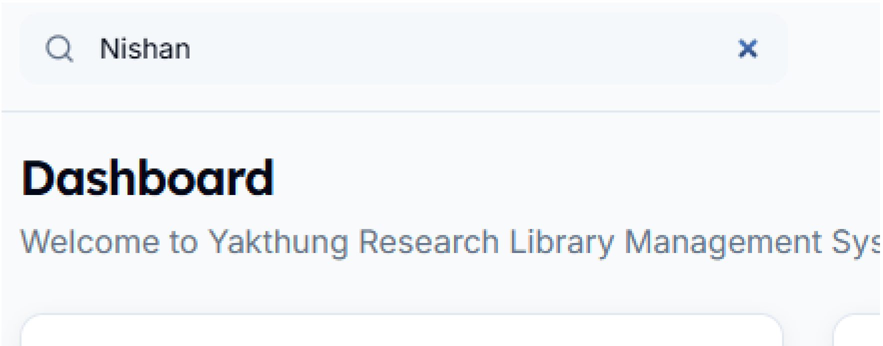
Test case 1. Verify that searching for a book by title returns the correct results.



Test case 2. Verify that searching for a book by author returns the correct results.



Test case 3. Verify that searching with no matching keywords returns an empty result set.

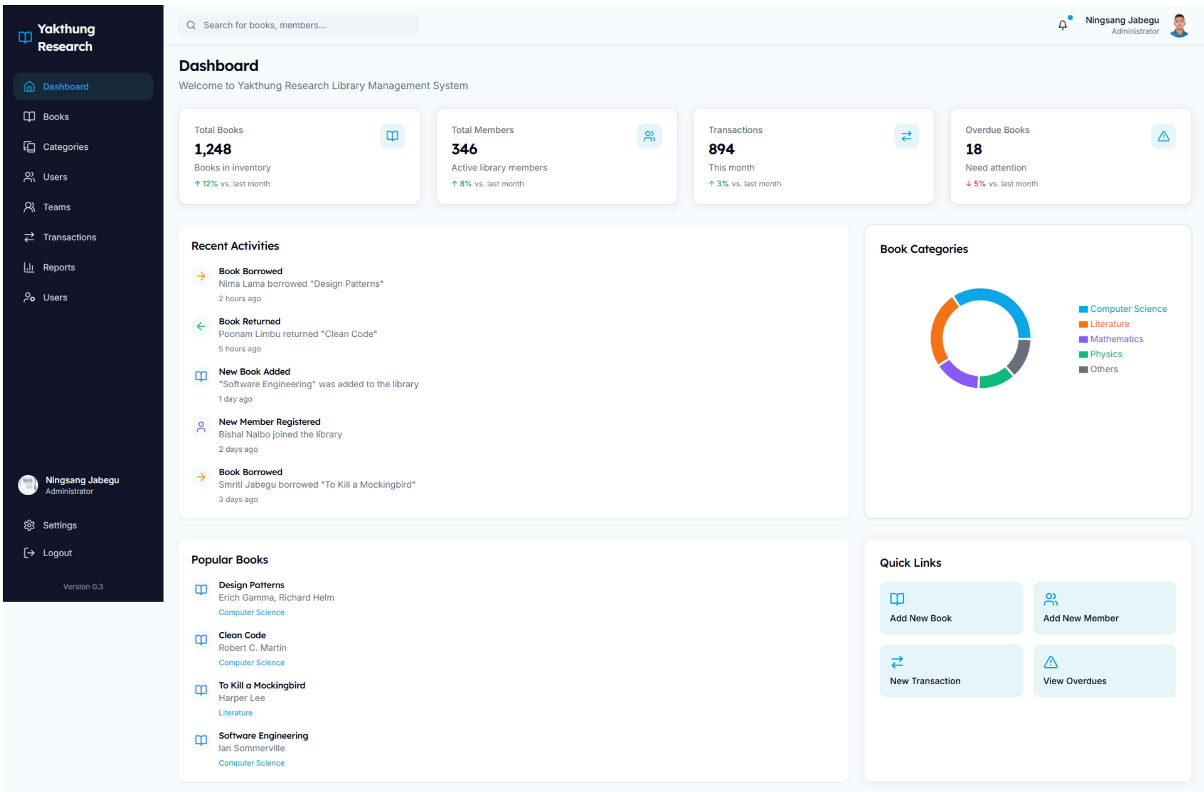


5.2.2. Test Cases for System Testing

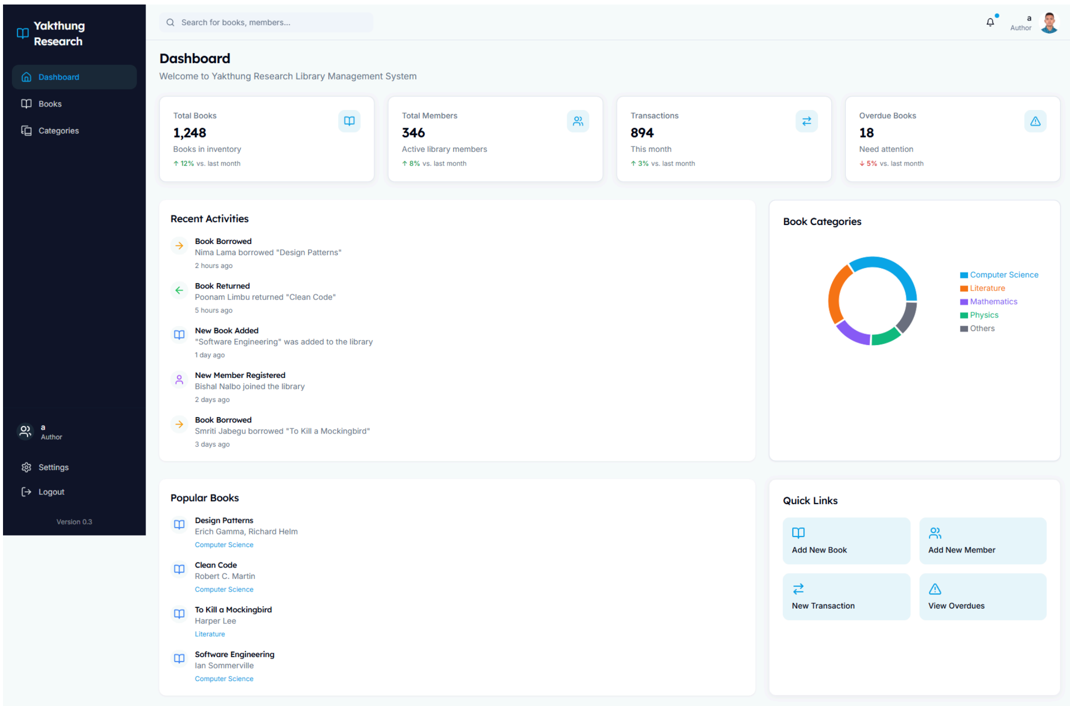
System testing was conducted to verify the overall functionality and integration of different modules of the system.[2] Examples of system test cases include:

Testing User Login and Role-Based Access.

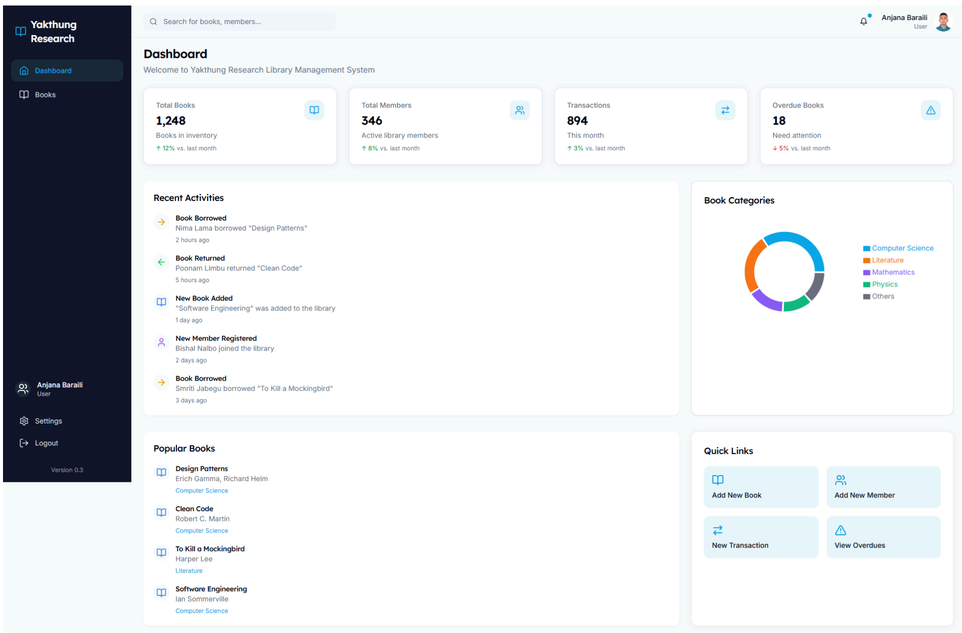
Test case 1. Verify that an administrator can log in and access all modules.



Test case 2. Verify that an author can log in and access only the allowed modules (Dashboard, Books, Categories).



Test case 3. Verify that a user can log in and access only the allowed modules (Dashboard, Books).



Testing Book Borrowing and Returning:

Test case 1. Verify that a logged-in user can successfully borrow an available book included in the appendices (Figure 5.5).

Test case 2. Verify that the borrowed book's status is updated to 'Checked Out' included in the appendices (Figure 5.6).

Test case 3. Verify that the same user can successfully return the borrowed book included in the appendices (Figure 5.5).

Test case 4. Verify that the returned book's status is updated to 'Available' included in the appendices (Figure 4.5).

Testing Report Generation.

Test case 1 Verify that the 'General Statistics' report displays accurate counts for total books, total members, checked-out books, and overdue books. A screenshot of the general statistics report will be included in the appendices (Figure 5.1).

Test case 2. Verify that the 'Book Categories Distribution' pie chart accurately reflects the number of books in each category. A screenshot of the book categories distribution chart will be included in the appendices (Figure 5.2).

5.3. Result Analysis

The unit testing phase resulted in a high percentage of passed test cases, indicating that the individual components of the system are functioning as expected. The system testing phase also showed that the integrated modules are working together correctly to provide the required functionalities. Minor bugs were identified during testing, primarily related to UI adjustments and edge-case scenarios in data validation. These bugs were promptly addressed and resolved by the development team. Overall, the testing results indicate that the Library Management System is robust and meets the specified functional and non-functional requirements.

6. Chapter 6: Conclusion and Future Recommendations

6.1. Conclusion

The project successfully developed a web-based Library Management System with the features and functionalities outlined in the user query. The system provides a user-friendly interface for managing library resources and operations, catering to different user roles including Administrator, Author, and User. Key functionalities such as user authentication, role-based dashboards, book and category management, transaction tracking, and report generation have been implemented and tested. The adoption of an Agile development methodology allowed for flexibility and iterative improvements throughout the project lifecycle. The system addresses the challenges associated with manual library management by automating key processes, thereby enhancing efficiency and providing a better experience for both library staff and patrons. The successful implementation of this project demonstrates the application of software engineering principles and best practices in developing a functional and practical web-based solution.

6.2. Future Recommendations

While the current version of the Library Management System (version 0.2) provides a comprehensive set of features, there are several potential enhancements and future work that could further improve the system:

6.2.1. Integration with External Library Databases

The system could be enhanced to integrate with external library databases or APIs to automatically fetch book information and metadata, reducing the manual effort required for cataloging.

Advanced Search Functionality. Implementing more advanced search options, such as searching by publication year, publisher, or subject keywords, could improve the discoverability of books for users.

Digital Resource Management. Future versions could include functionalities for managing digital resources such as e-books, journals, and other online materials.

Inter-Library Loan Management. Implementing a module to manage inter-library loan requests would allow users to borrow materials from other libraries.

Mobile Application Development. Developing a mobile application for the Library Management System would provide users with convenient access to library services on their smartphones and tablets.

Enhanced Reporting and Analytics. Adding more sophisticated reporting and analytics features, such as generating custom reports and visualizing library usage patterns, could provide valuable insights for library management.

Integration with Payment Gateways. For libraries that charge fines or membership fees, integrating with payment gateways would automate the payment process.

Accessibility Improvements. Further efforts could be made to enhance the system's accessibility for users with disabilities, ensuring compliance with WCAG guidelines.

These recommendations represent potential avenues for future development and could further enhance the capabilities and user experience of the Library Management System.

7. Bibliography (if any)

No additional sources were consulted beyond those listed in the References section.

8. Appendices

8.1. Appendix A: Title Page

TRIBHUVAN UNIVERSITY

INSTITUTE OF SCIENCE AND TECHNOLOGY

AMRIT SCIENCE CAMPUS

Department of Computer Science and Information Technology

Project Report: Development of a Web-Based Yakthung Research Library Management System

This is a project work report submitted in partial fulfillment of the requirements for the Bachelor of Science in Computer Science and Information Technology (BSc. CSIT) degree third year sixth semester.

2081-12-27

Submitted by:

Name: Ningsang Jabegu

Roll No.: 28198 / 078

TU Regd. No.: 5-2-33-67-2021

Batch : 2078

Course: Software Engineering (CSC 364)

Semester: VI

8.2. Appendix B: Declaration

Declaration

I, Ningsang Jabegu, declare that the project titled "Yakthung Research Library Management System" is my original work. I affirm that I developed this project independently under proper academic guidelines and professional ethics.

I extend my sincere gratitude to everyone who supported me during this project. I appreciate the insightful guidance received from my project supervisor, whose valuable feedback helped shape this work. I also thank the open-source communities behind the technologies I used, including React, Node.js, Express, and MongoDB. Their resources and discussions helped me overcome development challenges.

I acknowledge the invaluable support and resources provided by Tribhuvan University's Institute of Science and Technology. The access to quality academic materials and the opportunity to work on this project have greatly contributed to my learning journey.

I declare that every part of this project is my work and that I have credited all external sources appropriately. I commit to upholding the principles of academic integrity and professional responsibility.

Ningsang Jabegu (28198 / 078)

8.3. Appendix C: Supervisor's Recommendation

Supervisor's Recommendation

I recommend awarding a high evaluation to the Yakthung Research Library Management System project. The work demonstrates strong technical competence and a clear understanding of software engineering principles. The system effectively manages library operations through user authentication, role-based dashboards, comprehensive book management, transaction processing, and dynamic reporting. The candidate has applied proven development methodologies, showing precision in implementation and thoroughness in testing.

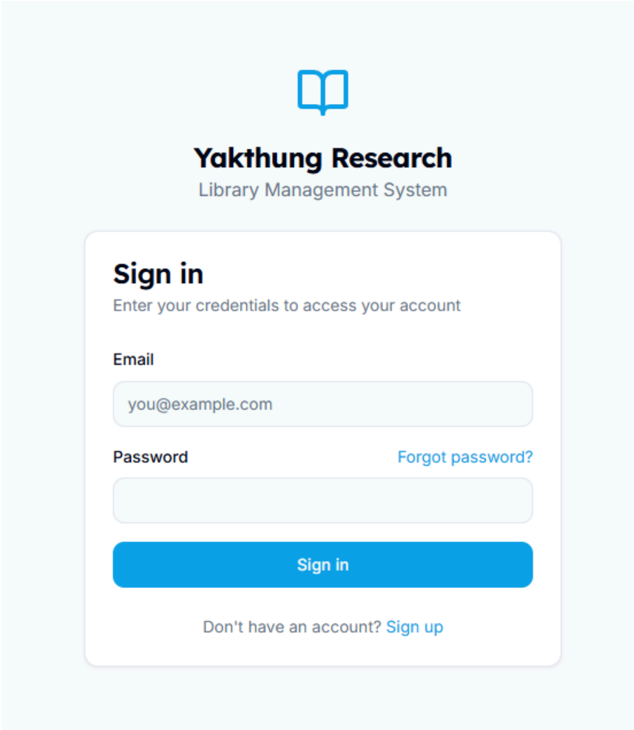
The project reflects exceptional problem-solving skills and an ability to manage complex requirements. The candidate's attention to detail is evident in the user interface design and seamless integration of back-end functionalities. The use of agile methodologies and systematic documentation further strengthens the project's quality. These aspects indicate both technical proficiency and a commitment to continuous improvement.

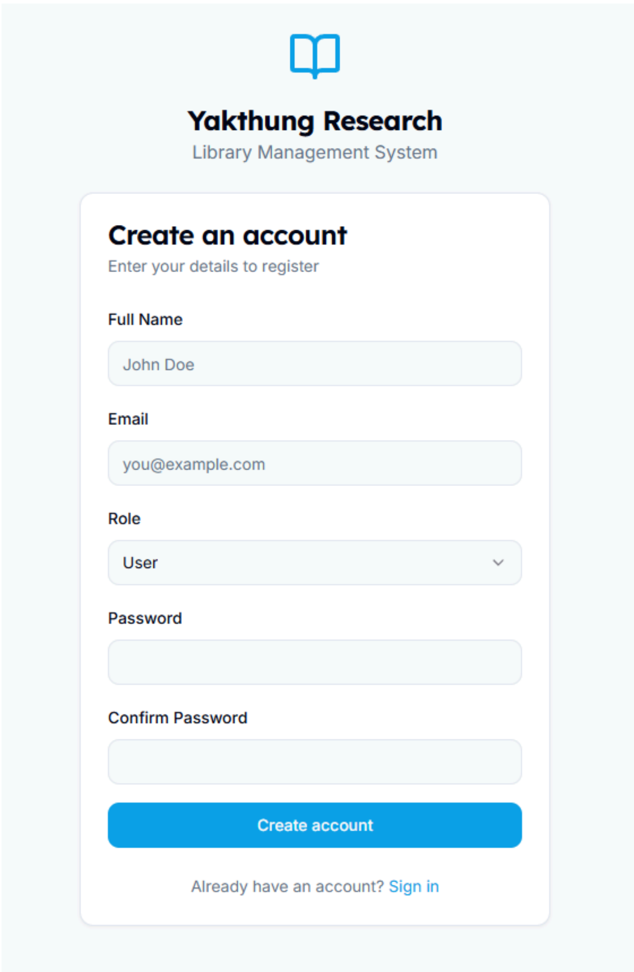
This project is a commendable example of a well-planned and executed system that meets practical needs in library management. I endorse the project as a solid contribution to the field of software engineering.

- **Asst. Prof. Chakra Rawal**

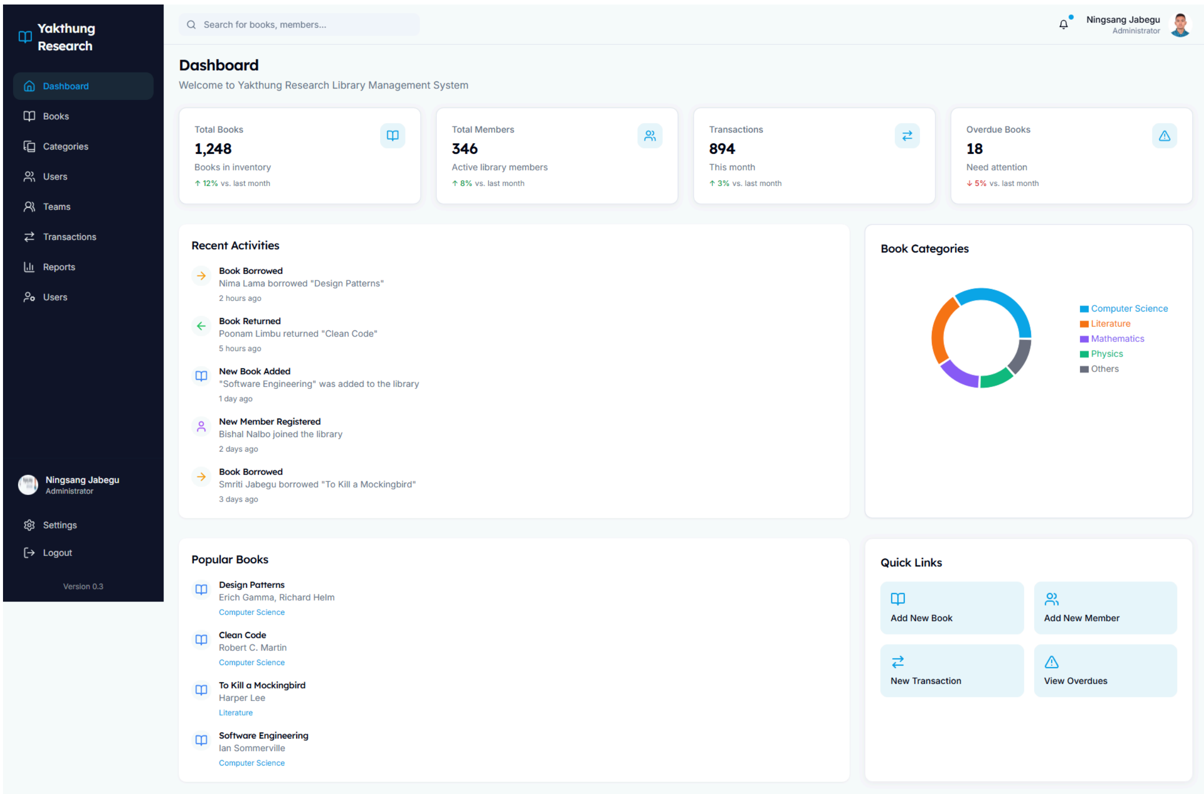
8.4. Appendix D: List of Figures

8.4.1. Figure 4.0: Sign In & Sign Up Page

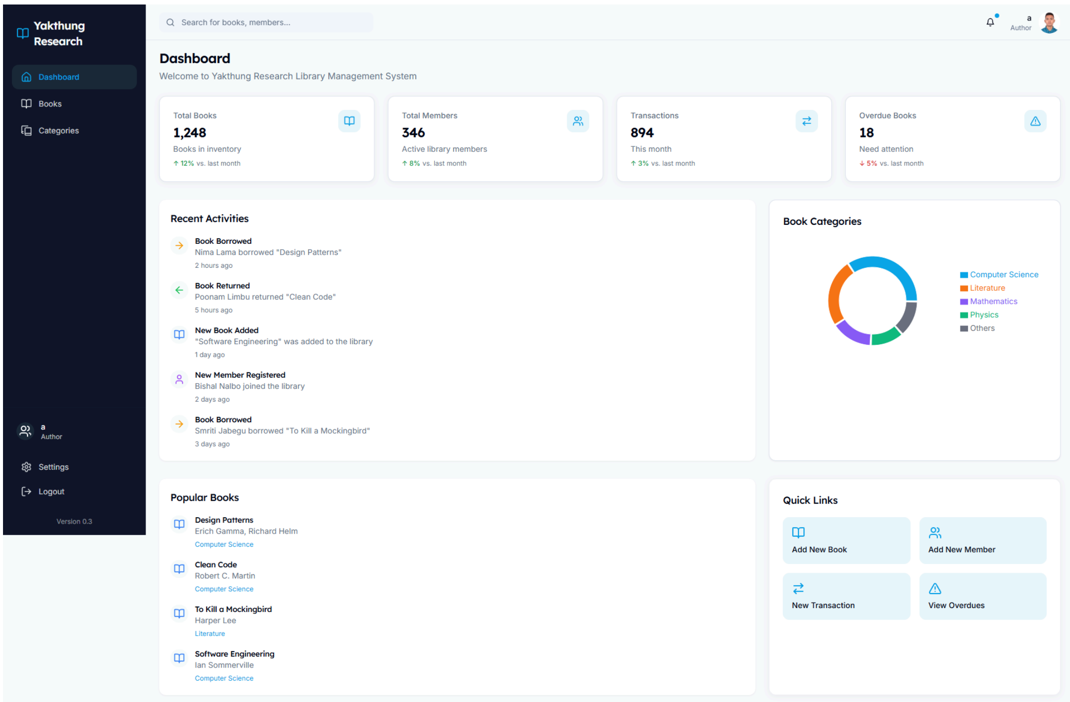

Yakthung Research
Library Management System
Sign in
Enter your credentials to access your account
Email
you@example.com
Password
Forgot password?
Sign in
Don't have an account? Sign up


Yakthung Research
Library Management System
Create an account
Enter your details to register
Full Name
John Doe
Email
you@example.com
Role
User
Password
Confirm Password
Create account
Already have an account? Sign in

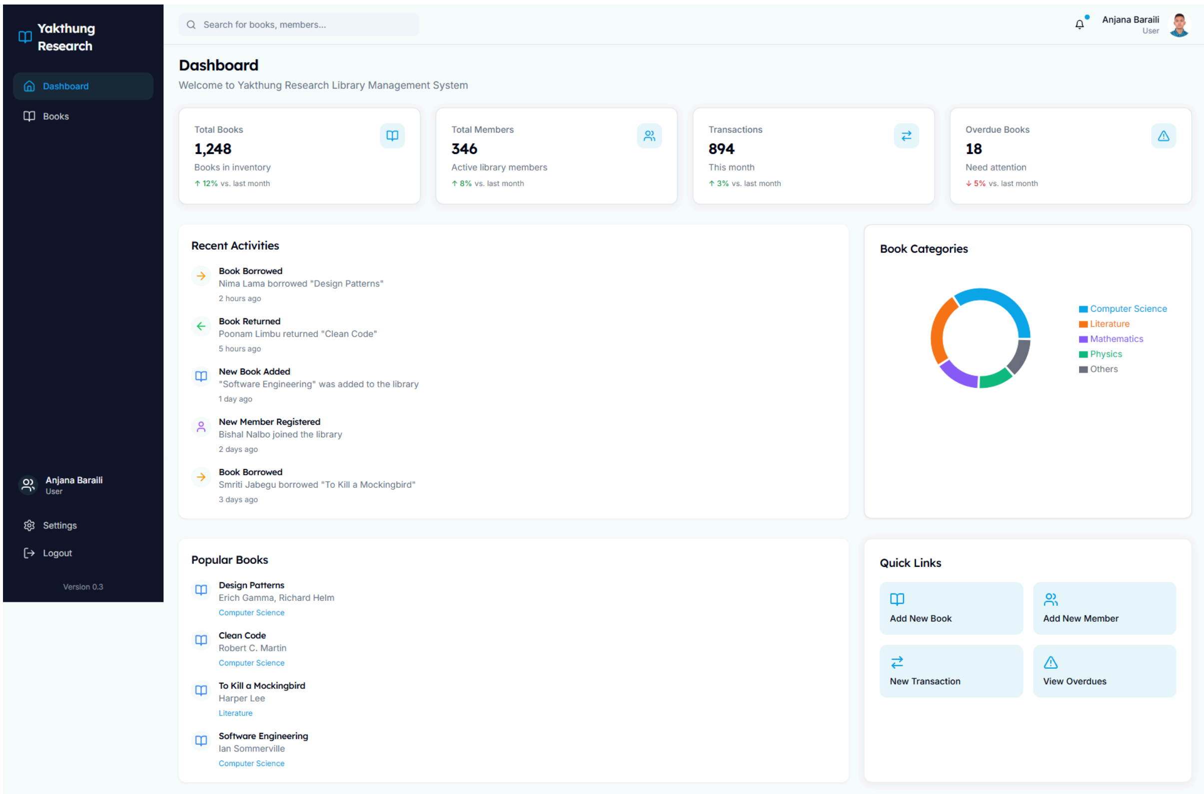
8.4.2. Figure 4.1: Administrator Dashboard



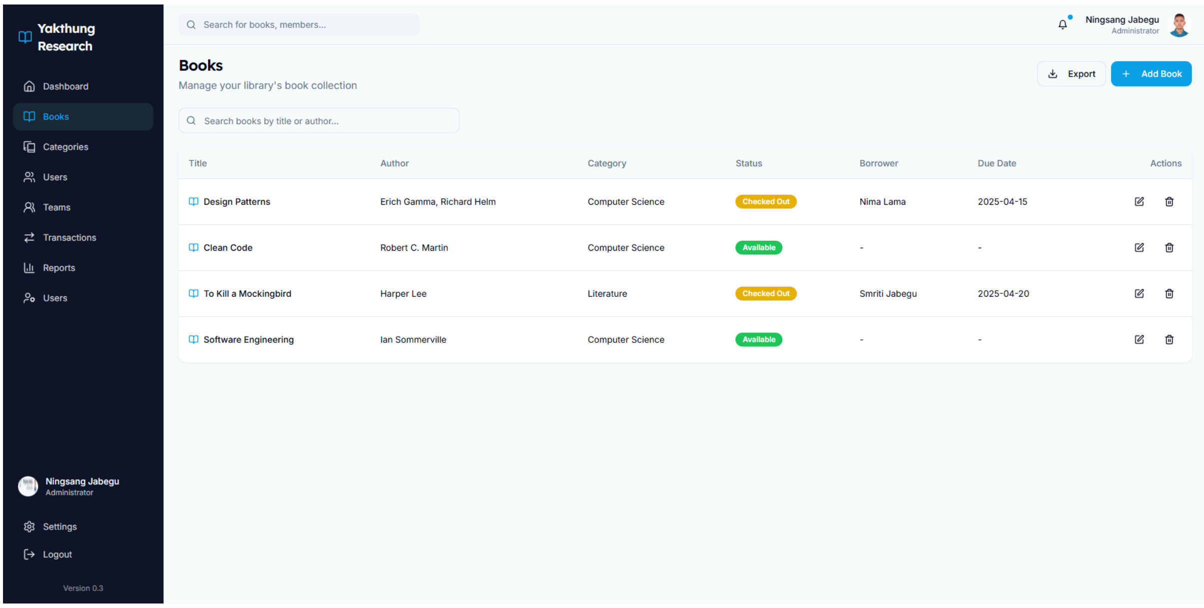
8.4.3. Figure 4.2: Author Dashboard



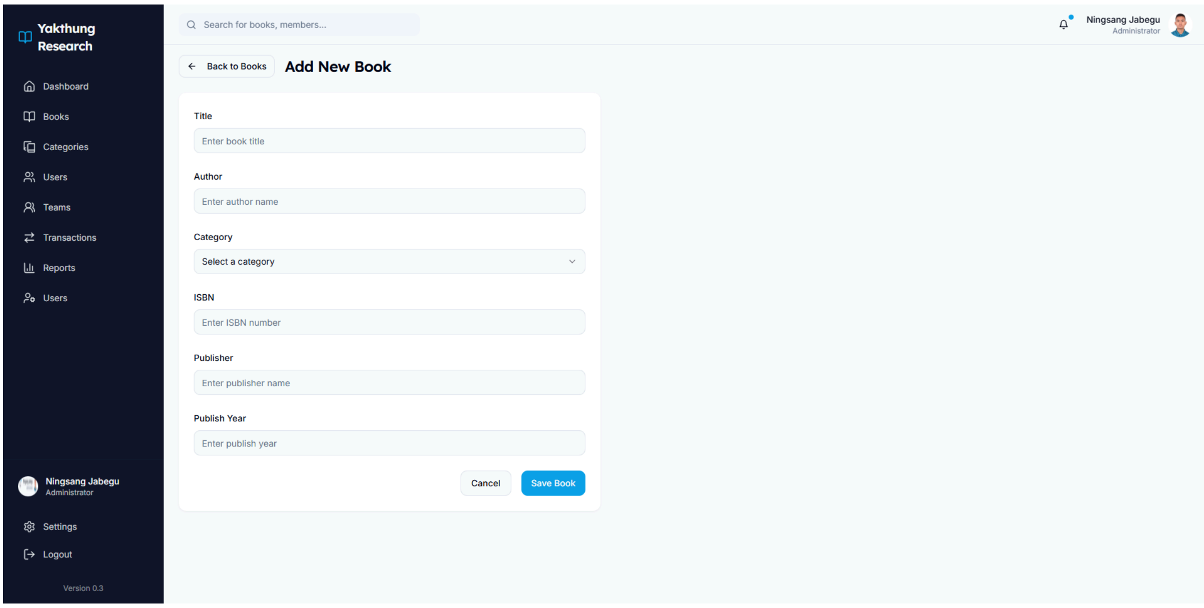
8.4.4. Figure 4.3: User Dashboard



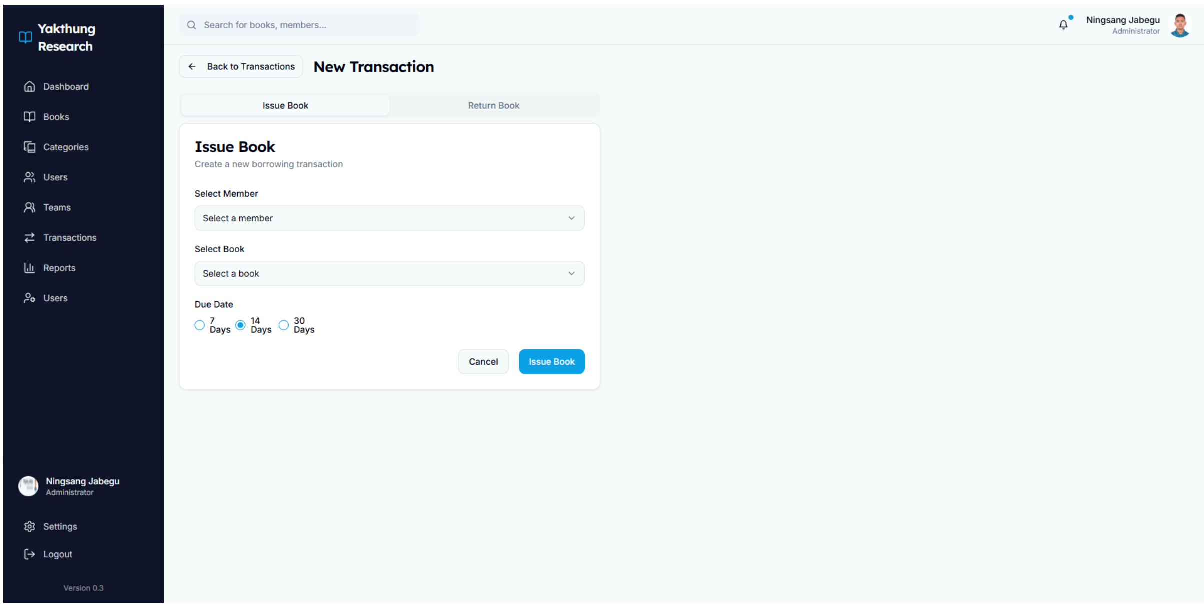
8.4.5. Figure 4.4: Book Listing Page



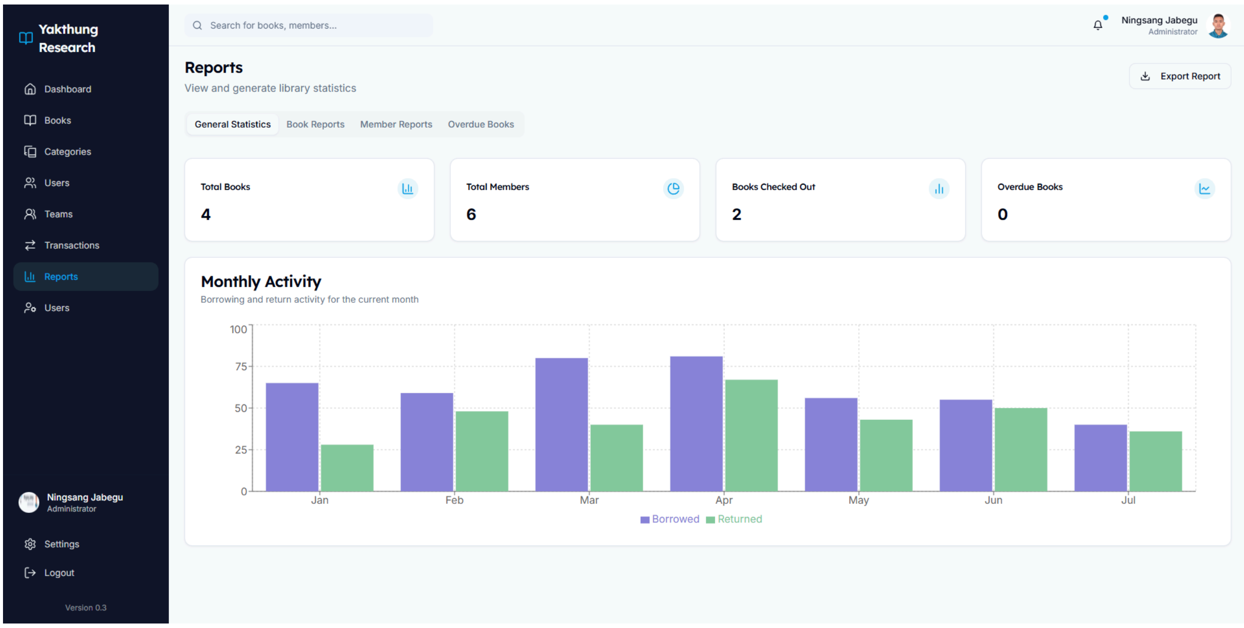
8.4.6. Figure 4.5: Add Book Form



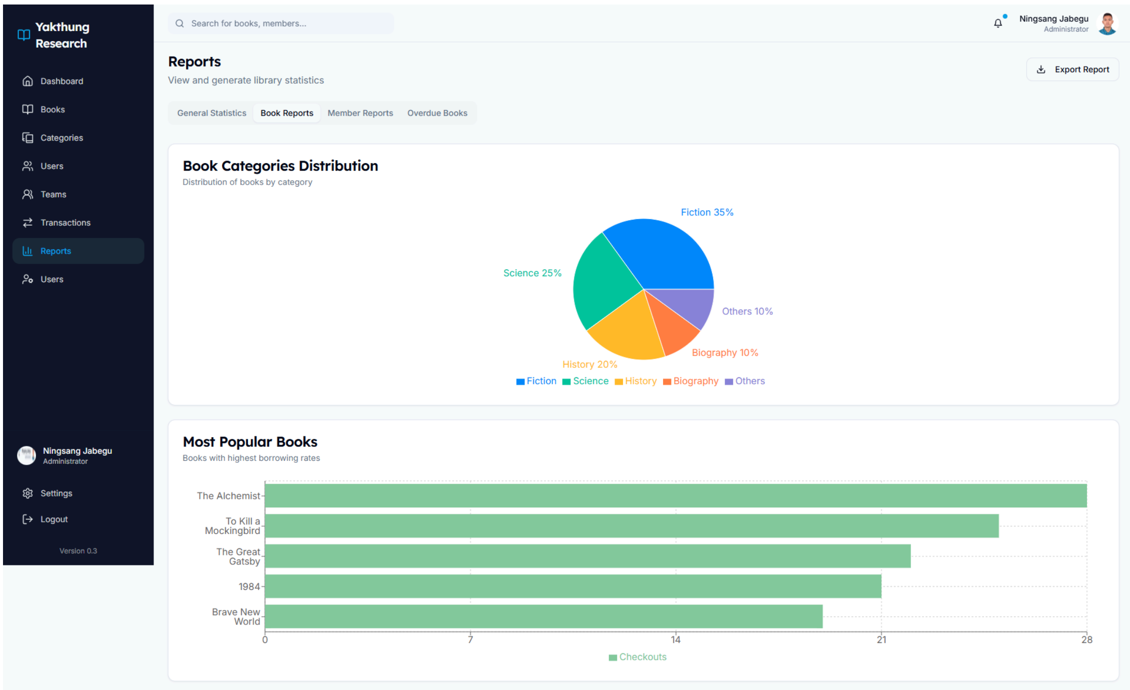
8.4.7. *Figure 4.6: Transaction Form*



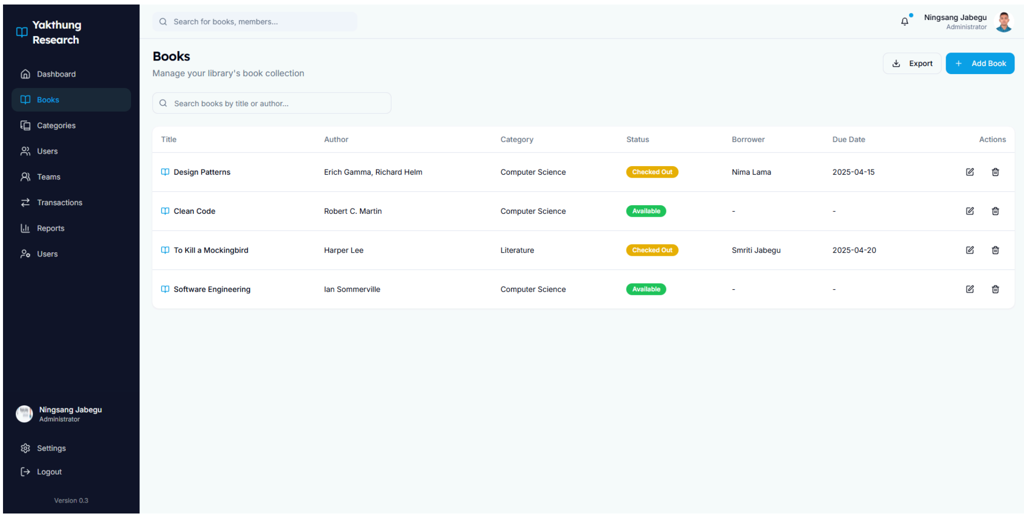
8.4.8. *Figure 4.7: General Statistics Report*



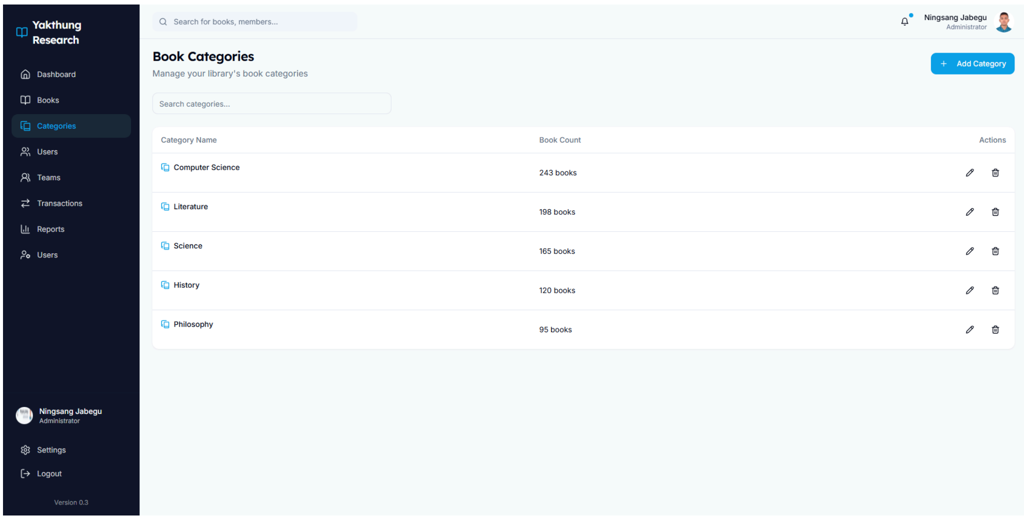
8.4.9. *Figure 4.8: Book Categories Distribution Chart*



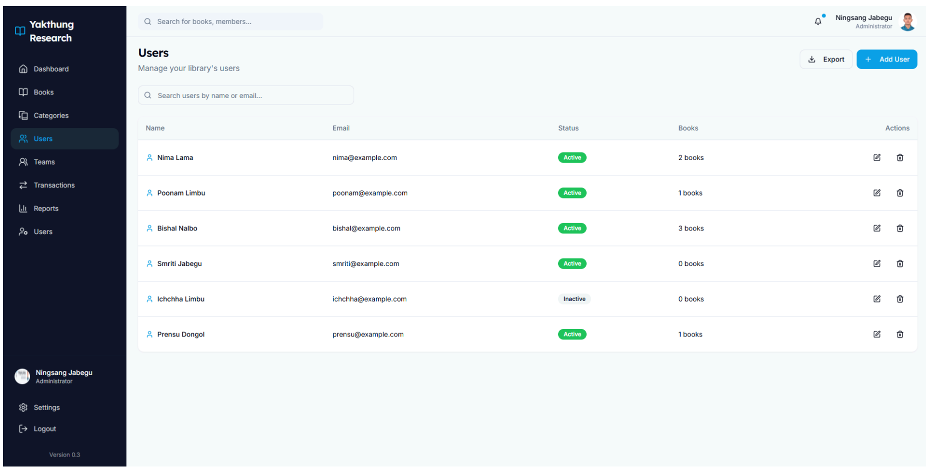
8.4.10. *Figure 5.1: Books Collection Page*



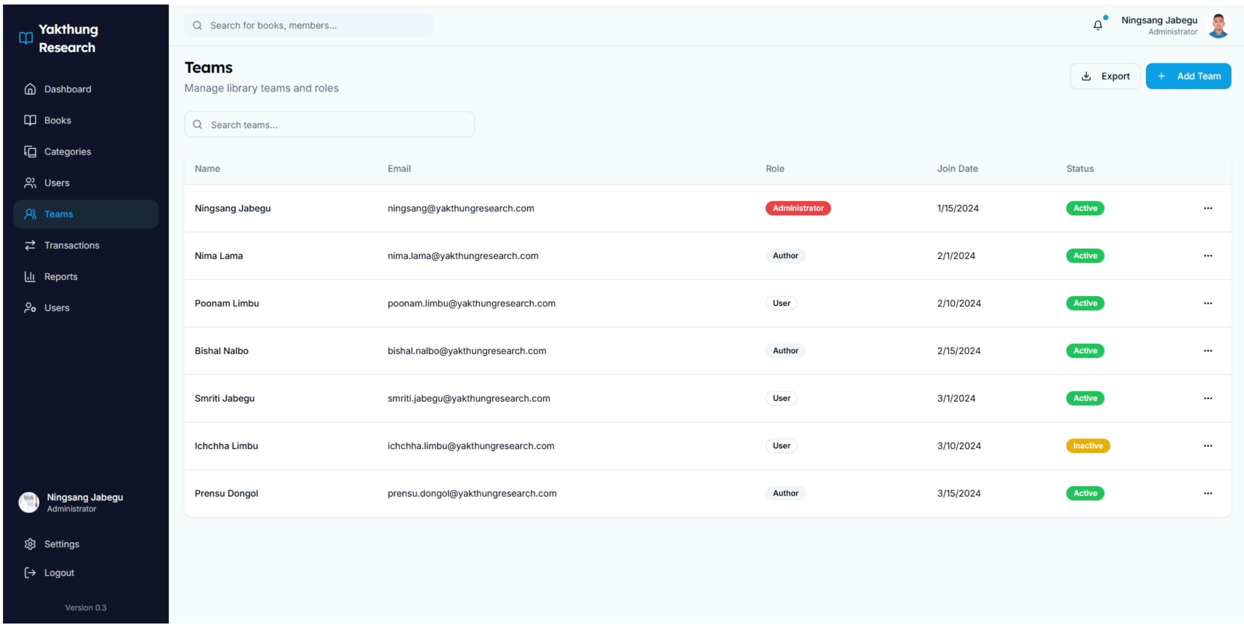
8.4.11. *Figure 5.2: Categories Collection Page*



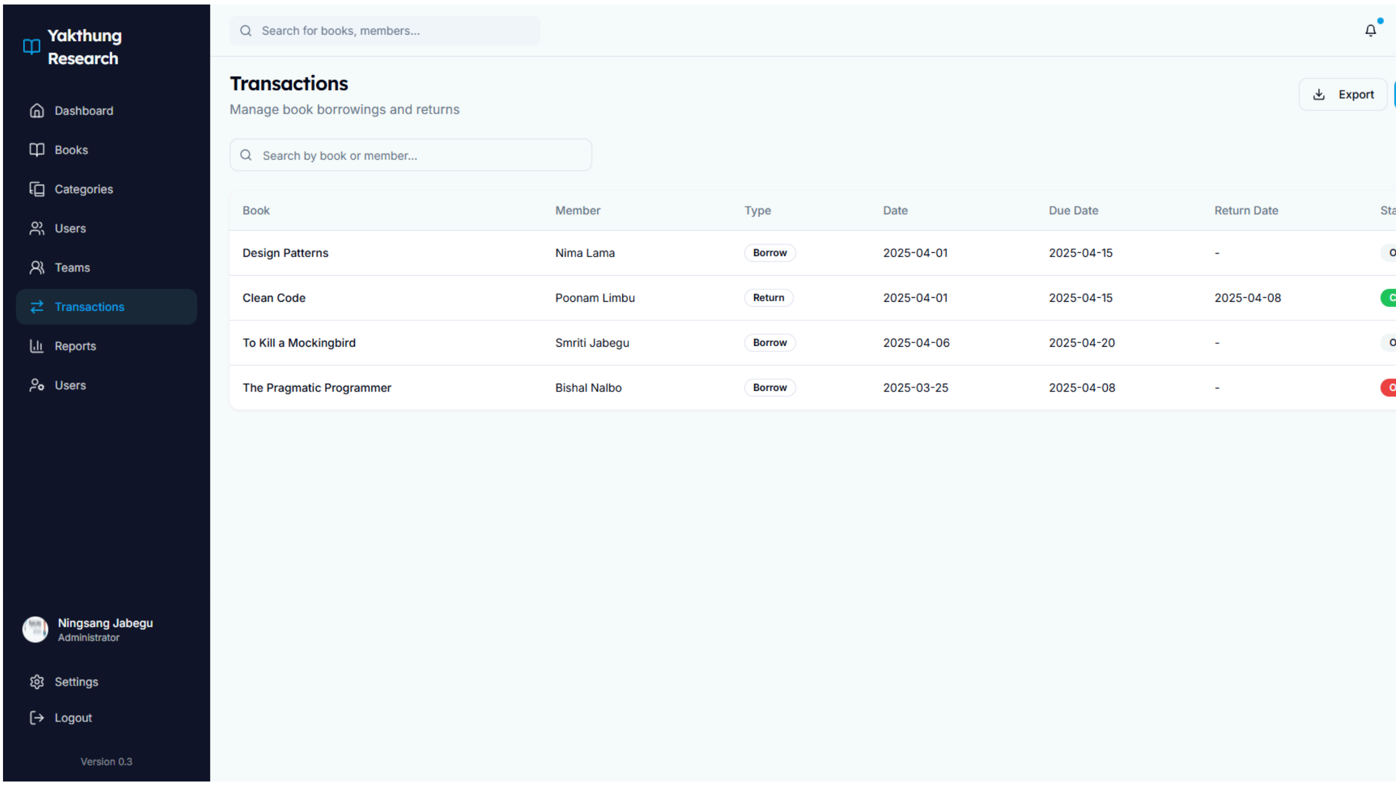
8.4.12. *Figure 5.3: Library's Users Page*



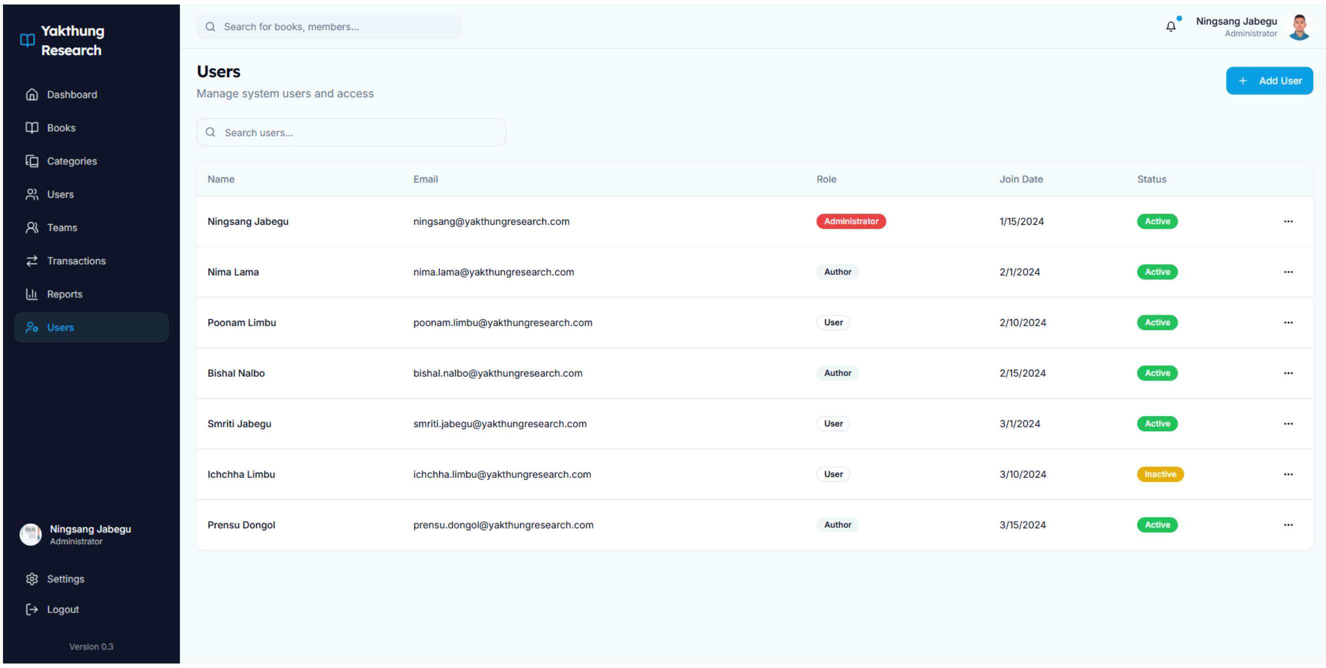
8.4.13. *Figure 5.4: Library's Teams Page*



8.4.14. *Figure 5.5: Library's Transactions Page*



8.4.15. *Figure 5.6: System Users and Access Page*


Yakthung Research
Dashboard
Books
Categories
Users
Teams
Transactions
Reports
Users
Ningsang Jabegu
Administrator
Settings
Logout
Version 0.3
Search for books, members...
Ningsang Jabegu
Administrator
Users
Manage system users and access
+ Add User
Search users...
Name
Email
Role
Join Date
Status
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